Code No: 155BT **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
IMAGE PROCESSING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Briefly state the digital image processing problems and applications.
b) With a suitable example, write the procedure for converting gray level image into a binary image. [7+8]

2.a) What is image interpolation? How to apply it for image resizing – explain?
b) Describe mechanisms of spatial filtering. [7+8]

3. Explain about smoothing spatial filters with suitable mathematical examples. [15]

4.a) Discuss about some important noise probability density functions.
b) Describe the procedure for restoration in the presence of noise only-spatial filtering. [7+8]

5.a) Write about Wiener filtering with example.
b) Write about constrained least square filtering. [7+8]

6. "Image thresholding enjoys a central position in allocation of image segmentation" – Justify the above fact with appropriate example. [15]

7.a) Discuss about image compression models in brief.
b) What is JBIG2 compression? Give example. [8+7]

8. Write short notes on the following.
a) DFT, DCT 2D transformations.
b) Interactive restoration. [8+7]

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